Setup

This tab explains how to set up your environment and tooling so you can run SWEArena sessions in a code editor using `zed-eval`.

Platform Prerequisites

Linux

Install docker on your linux distro.

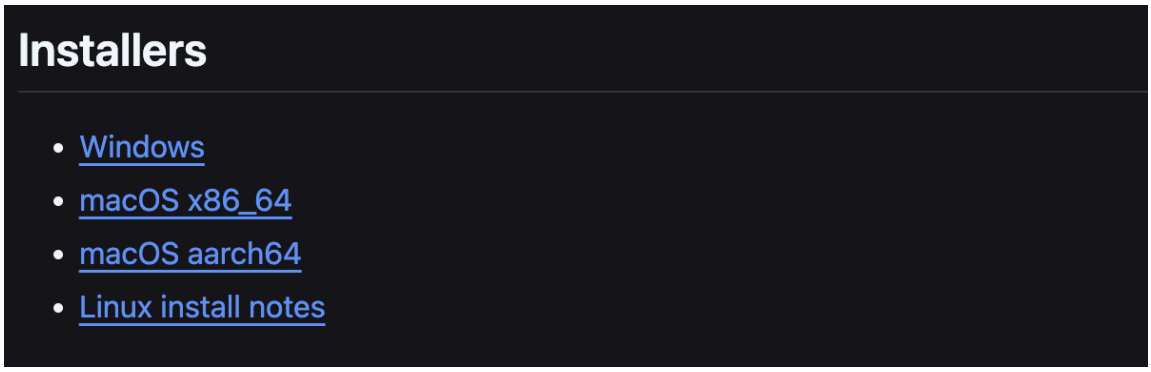
Windows: install WSL (required) and Rancher Desktop

Install git with Git Bash for Windows.

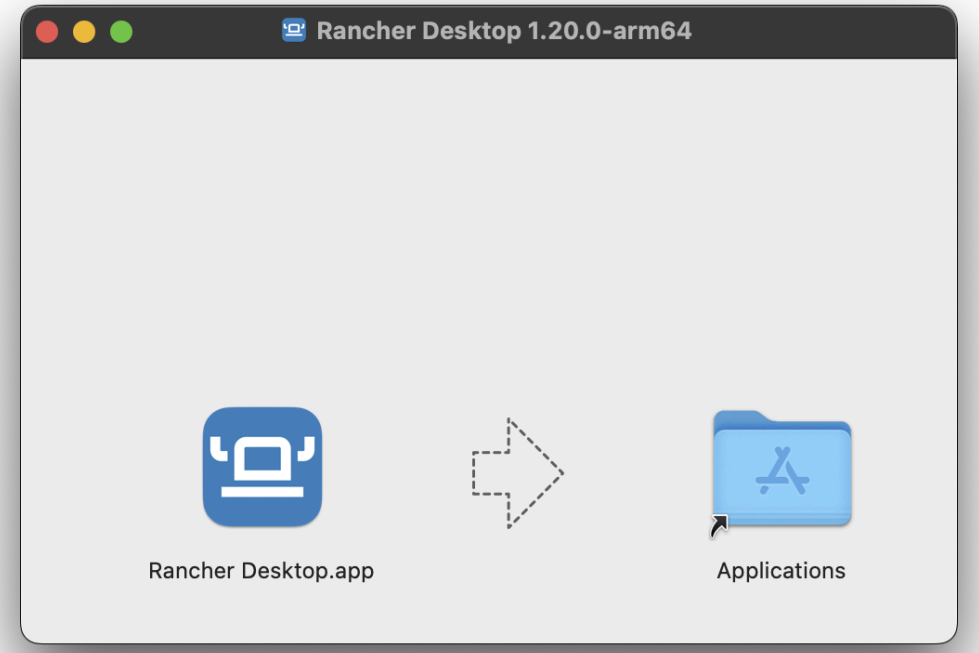
Please watch the following video for WSL + Rancher Desktop + zed-eval script setup:
WSL_Setup.mp4

Mac: Install Rancher Desktop

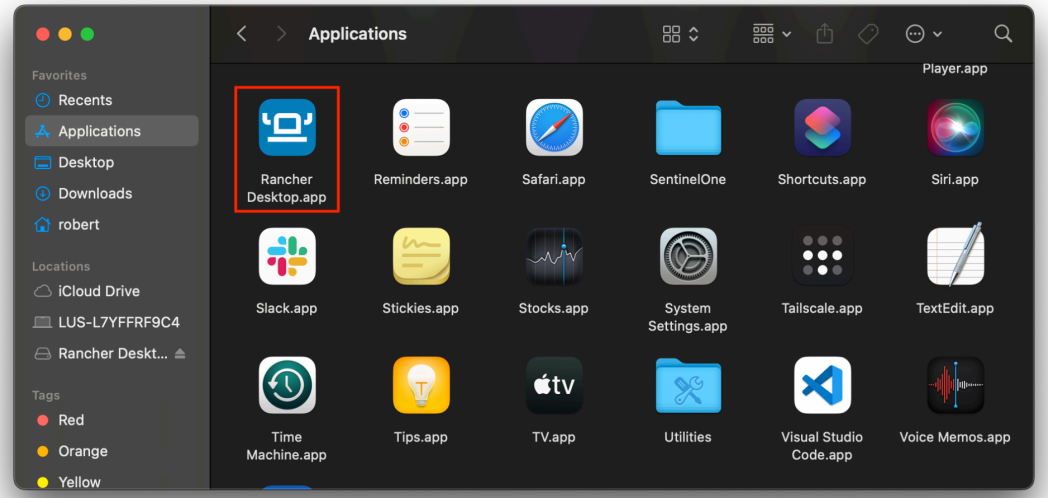
1. Download the latest release for your processor architecture from the Rancher Desktop Github Releases page. For the *M series Apple silicon*, use *aarch64*, otherwise use *X86_64*.



2. Open the Downloaded DMG image file and drag the *Rancher Desktop* app to your *Applications* folder. *Rancher Desktop* will install.

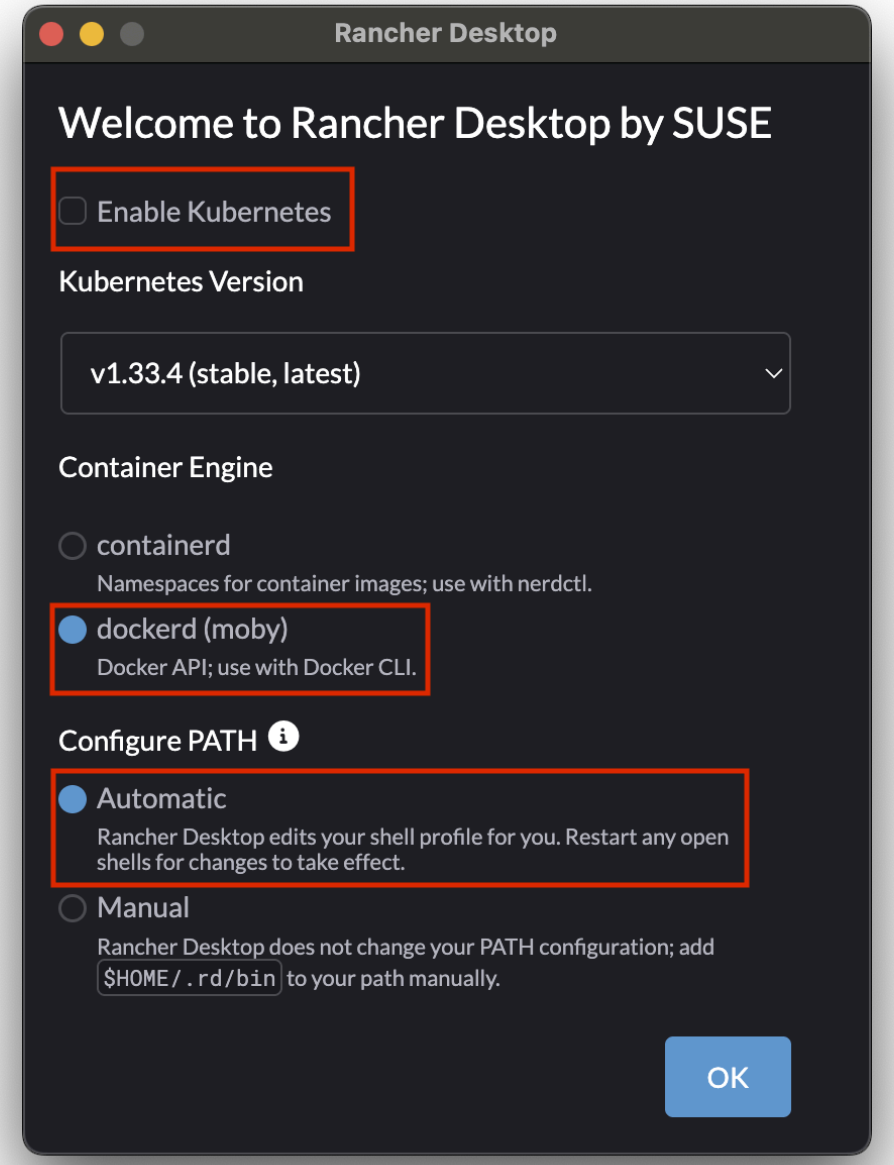


3. After installation, open *Rancher Desktop* from your Applications/Launchpad.



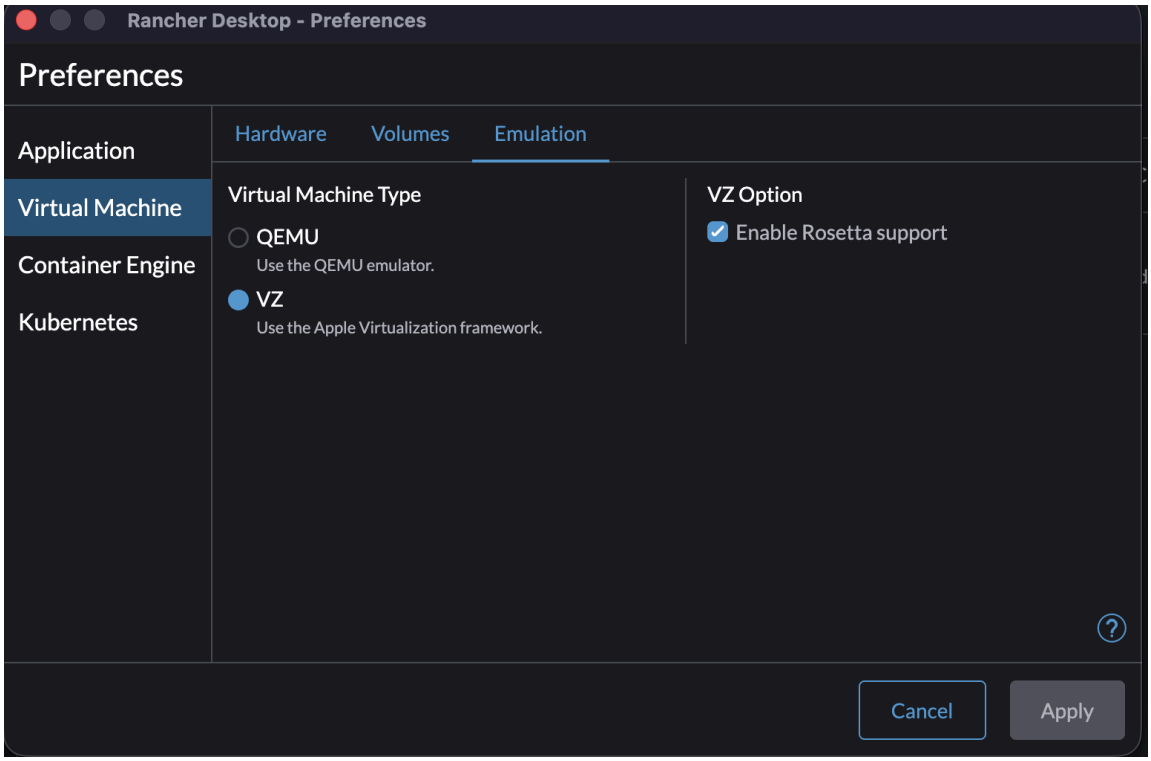
4. Do not enable *Kubernetes* - we currently do not make use of it and enabling it has caused issues, historically. Make sure the container engine is *dockerd(moby)* and the *PATH* configuration is set to *automatic* - unless you want to set up your *PATH*

environment variable

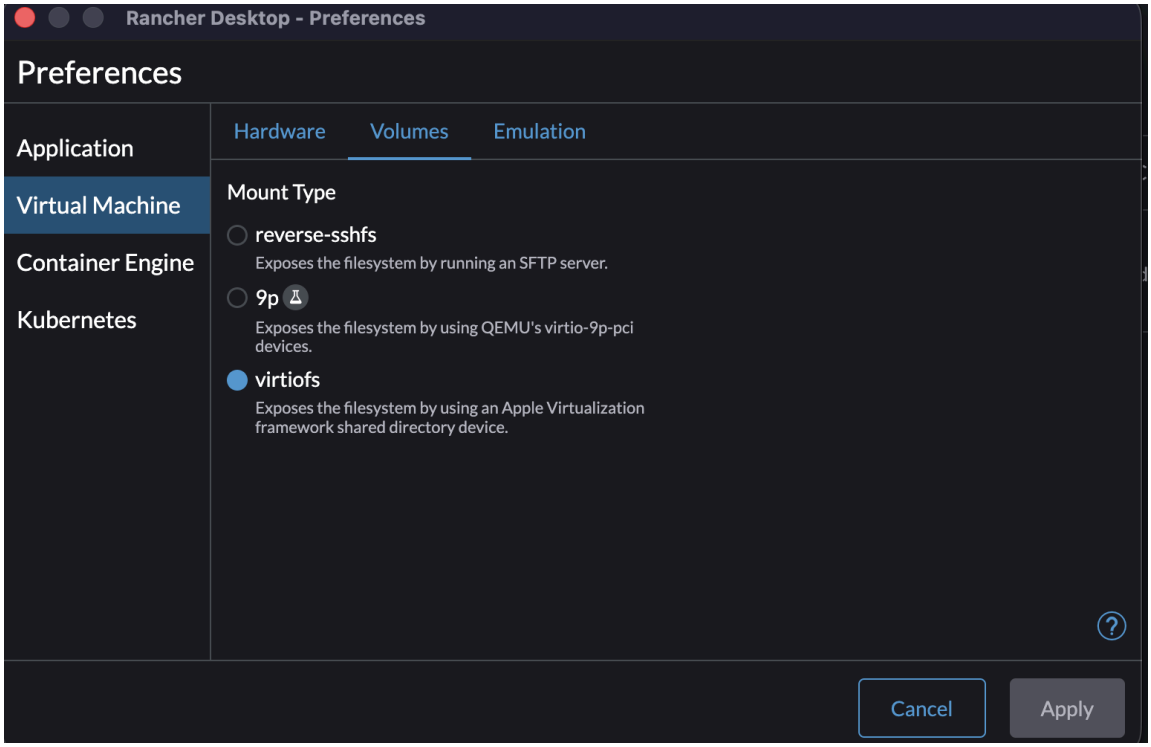


Enable Linux container support

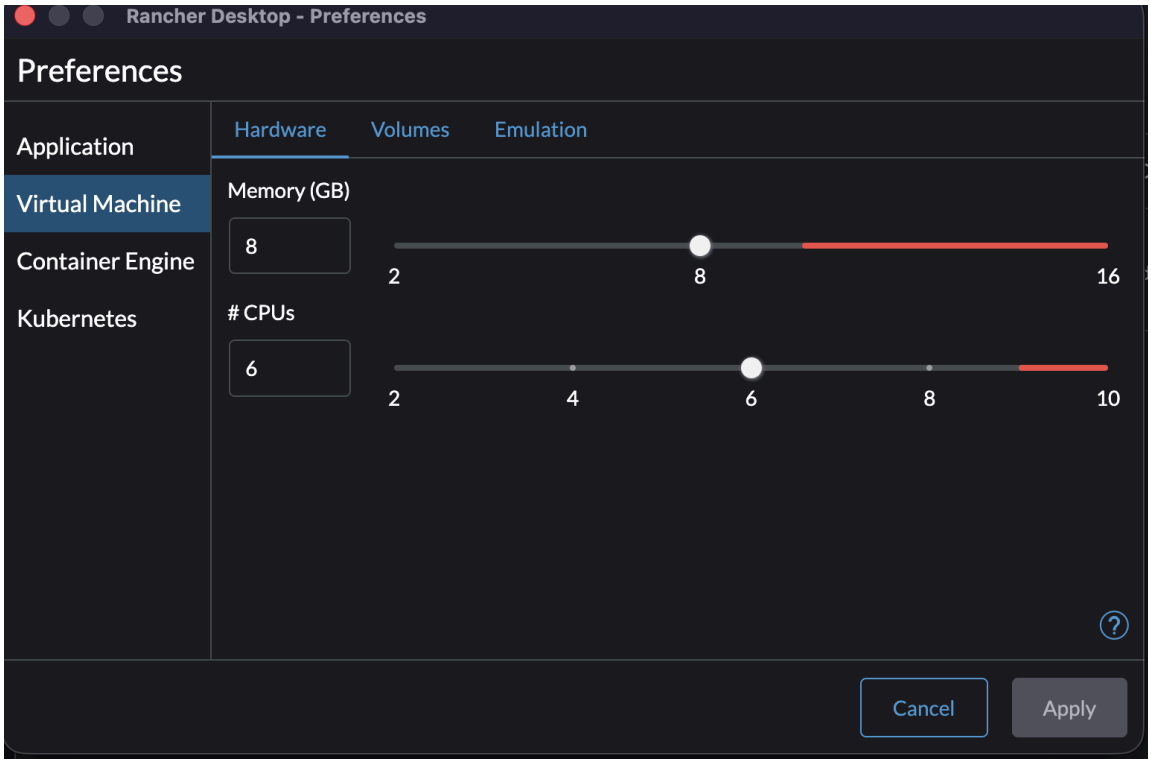
- Grok TUI build version >= 0.1.100 support amd64 with rosetta


Rancher Desktop - Preferences
Preferences
Application
Virtual Machine
Container Engine
Kubernetes
Hardware
Volumes
Emulation
Virtual Machine Type
QEMU
Use the QEMU emulator.
VZ
Use the Apple Virtualization framework.
VZ Option
Enable Rosetta support
Cancel
Apply

-


Rancher Desktop - Preferences
Preferences
Application
Virtual Machine
Container Engine
Kubernetes
Hardware
Volumes
Emulation
Mount Type
reverse-sshfs
Exposes the filesystem by running an SFTP server.
9p
Exposes the filesystem by using QEMU's virtio-9p-pci devices.
virtiofs
Exposes the filesystem by using an Apple Virtualization framework shared directory device.
Cancel
Apply

-

- 

- Allocate as much resource your system allows for VM memory and CPU
- Make sure Docker is running and run:

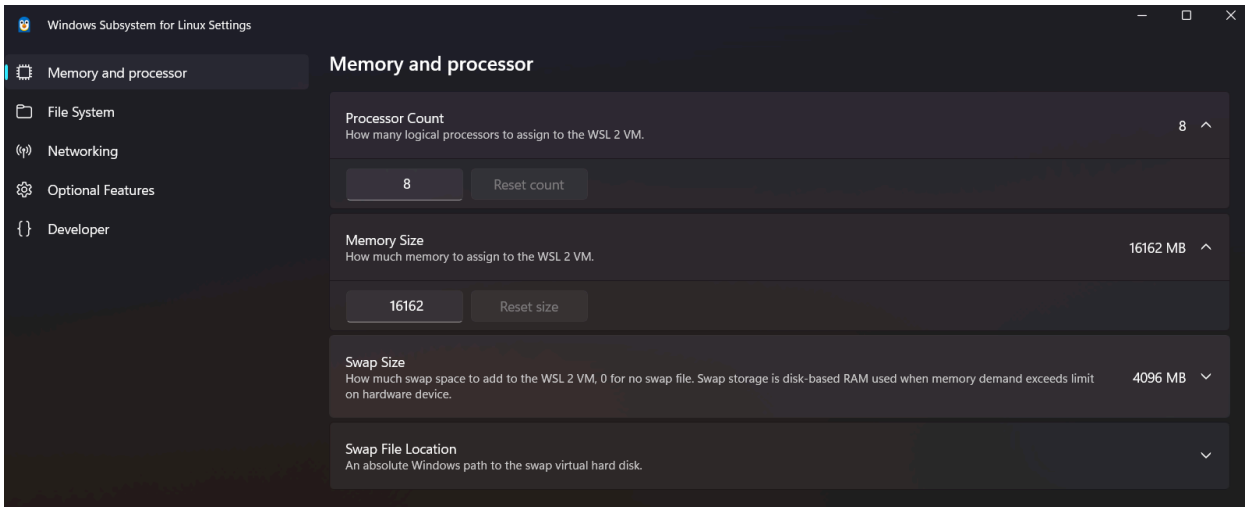
****IF Your system has docker desktop installed , please uninstall it or at the very minimum force quit it and run the following command****

```shell
docker context use rancher-desktop
```

Allocating Resources to containers:

WSL:

- Search for WSL Settings in the task bar and open it.
- Under Memory and processor select a minimum of 4 cores and 8192 MB of RAM



Linux/MacOS:

- Allocate as much memory and CPUs as you can. A minimum of 8GB of memory and 4 CPUs is recommended.

1. Get the `zed-eval` package

- If you have access: clone `https://nova.teachx.ai/qwakeley/zed-eval`
- Otherwise: download the zip from zed-eval **(pinned in #hd-dev-arena-pilot)**

2. Configure `zed-eval`

Install dependencies

1. Install git
2. Navigate to your zed-eval folder

3. Install `uv` (Python): `https://docs.astral.sh/uv/getting-started/installation/`
4. Install dependencies:
   - `uv sync`

Create `config.toml`

Create a file in your zed-eval directory called `config.toml`:

```
None
[ui]

[cli]
installer = "npm"
auto_update = true

[models]
default = "grok-code-fast-1"

[model.codex]
model = "gpt-5.2-codex"
base_url = "https://research-models.api.x.ai/research/requesty"
name = "GPT 5.2 Codex"
description = "GPT 5.2 Codex via Requesty"
temperature = 1
api_key =
"xai-JECkm6NijuM1OFwvhUwF0Wgdhp7dtZH23YZHVURkQohKnXXok0jKVY1O6PObasqs9rz5wrzUyg
ZsUPXl"
context_window = 400_000

[toolset]
use_concise = true

[harness]
enable_upload_logging = true
block_for_upload = true
disable_codebase_upload = true
```

Create `.env`

Create a `.env` file in your zed-eval directory with the following values. Make sure to replace **TUTOR_NAME** with your name!

```
None
NPM_AUTH_TOKEN=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
SSH_PORT=2222
TUTOR_NAME="[FirstNameLastName]"
PROJECT_ID="p-WSpCWGptzH60ooRHZgs69"
```

GCloud Setup

Note: If you're using WSL/Ubuntu use the Debian/Ubuntu section

1. Install GCloud cli (don't create an account)
2. `gcloud auth login` and sign in with your **teachx google account**
3. `gcloud auth configure-docker asia-south1-docker.pkg.dev` **(make sure to press 'y')**
4. Verify you can access our registry (run this from the zed-eval folder):
   a. `gcloud artifacts repositories describe hd-swe-swearena --location=asia-south1 --project=xai-human-data-infra`
5. **If and only if you are still facing docker auth issues, try this:**
   a. `docker login asia-south1-docker.pkg.dev/xai-human-data-infra/hd-swe-swearena`
   b. Make sure REGISTRY_USERNAME and REGISTRY_PASSWORD aren't set in .env

If you get this error: `IAM_PERMISSION_DENIED`, please ask a team lead to grant you access.

Output of **step 4** should look similar to this:

```
None
Encryption: Google-managed key
Repository Size: 1037.998MB
cleanupPolicyDryRun: true
createTime: '2026-01-21T00:31:49.077612Z'
description: For India tutors
dockerConfig: {}
format: DOCKER
mode: STANDARD_REPOSITORY
name:
projects/xai-human-data-infra/locations/asia-south1/repositories/hd-swe-swearen
a
registryUri: asia-south1-docker.pkg.dev/xai-human-data-infra/hd-swe-swearena
```

```
satisfiesPzi: true
satisfiesPzs: true
updateTime: '2026-01-21T04:04:41.868072Z'
vulnerabilityScanningConfig:
  enablementConfig: DISABLED
  enablementState: SCANNING_DISABLED
  enablementStateReason: Vulnerability scanning is disabled.
```

3. Create a base `Dockerfile` (for `zed-eval`)

Update the `Dockerfile` inside `zed-eval` to represent the **initial state** of your task environment (the state at the first turn).

Guidelines:

- Keep it minimal: only what's required to run the repository.
- Replace the REPO_URL and COMMIT_HASH with your chosen public github repository.
- Pin external packages/versions (apk/apt/pip/etc.).
- Set the workdir to `/testbed`. (`WORKDIR /testbed/${REPO_NAME}`)
- Clone the repository.
    - For projects building on top of a public Github repository, this is the link to your Github repo, eg: https://github.com/apache/kafka
    - For `0->1` prompts, please clone this repo: `https://github.com/qwakeley-teachx/zed-base`
- (optional, for big repos) Pre-compile the repository in the dockerfile.

Tip: If you're stuck, copy the example Dockerfile in `zed-eval` and adapt it.

4. Run `zed-eval`

1. Run:

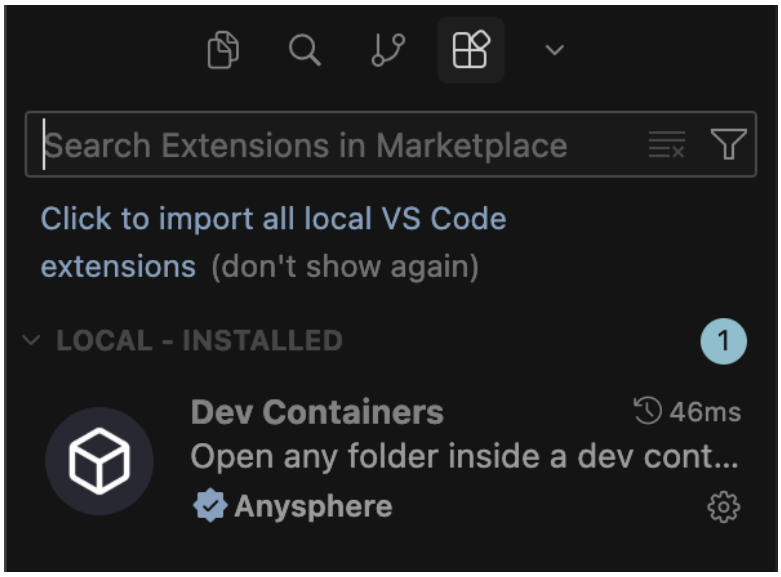
```shell
uv run zed-eval
```

What this does: builds your Dockerfile, creates the container, installs grok installation for you.

5. Attach to running container

Ensure you have either [VSCode]() or [Cursor]() installed and use the respective command to attach the container.

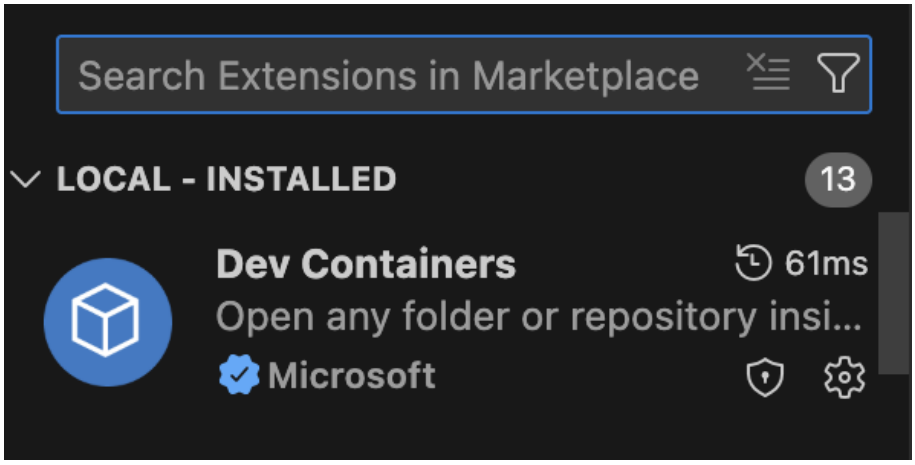
A) If you have Cursor installed make sure you have the extension `devcontainers` installed



Run the following command to attach to the container

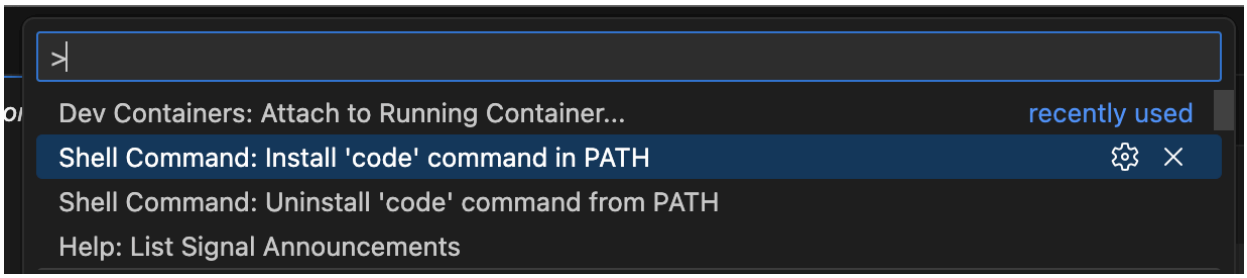
```shell
uv run zed-cursor
```

B) If you have VS code installed make sure you have the extension `devcontainers` installed



Run the following command to attach to the container

First please ensure `code` is available in the path



Verify its accessible with `which code`

```shell
uv run zed-vscode
```

WSL issues see: this [thread]

Next you need to authenticate grok, after attaching to the container run the following command in a terminal:

```
grok login
```

This will give you an URL which you need to copy and paste in the browser.

Login with your teachx account and it will give you a code, come back to the terminal and paste the code.

```
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
grok@be247650590d:~$ grok
Grok Build - v0.1.85 /home/grok

----------------------------------------------------------------[welcome]----------------------------------------------------------------

  Visit this URL to get your auth token:
  https://accounts.x.ai/sign-in?redirect=grok-cli&return_to=http://localhost:42455/on_token

  Enter token: |
  (Press Enter to submit, Esc to quit)
-----------------------------------------------------------------------------------------------------------------------------------------
  [^d,^q]=quit [^c]=clear [↵]=select
```

Once you are authenticated, run

```
None
XAI_USER=1 XAI_ROOT=1 grok --yolo --model grok-code-fast-1
```

After initially prompting the agent, ensure `zed-eval` is correctly picking up changes. Your logs should look similar to the following for each turn with the agent.

```
TURN 1 IN PROGRESS

>>> Press 'q' or Ctrl+C to FINISH and UPLOAD to registry <<<

2026-01-17 19:09:41.024 | INFO     | zed_eval.eval:watch_for_new_prompt:88 - New prompt detected (entries: 1 from 0)
sha256:11dc720b54c506c58e90c46488120a9a0a067973559bb58b43933653a6267090
2026-01-17 19:09:43.314 | INFO     | zed_eval.docker:commit_image:282 - ✓ Committed: zed-eval-session:QuitmanWakeley_test__16f368e6_1_fix_order

✓ TURN 1 COMMITTED

Image: zed-eval-session:QuitmanWakeley_test__16f368e6_1_fix_order

TURN 2 IN PROGRESS

>>> Press 'q' or Ctrl+C to FINISH and UPLOAD to registry <<<

2026-01-17 19:09:46.224 | INFO     | zed_eval.eval:watch_for_new_prompt:88 - New prompt detected (entries: 2 from 1)
sha256:a44929f49b83bcc19beb6c7498151d05bdf9d136de33c37926ff30a1719c6805
2026-01-17 19:09:48.173 | INFO     | zed_eval.docker:commit_image:282 - ✓ Committed: zed-eval-session:QuitmanWakeley_test__16f368e6_2_fix_order
2026-01-17 19:09:48.608 | INFO     | zed_eval.eval:watch_for_new_prompt:88 - New prompt detected (entries: 3 from 2)

✓ TURN 2 COMMITTED

Image: zed-eval-session:QuitmanWakeley_test__16f368e6_2_fix_order

TURN 3 IN PROGRESS

>>> Press 'q' or Ctrl+C to FINISH and UPLOAD to registry <<<

sha256:8ac21e312bcd8f38c3595b7275ce488e8f5f32ddba2b1718e3c83835340dd349
2026-01-17 19:09:50.574 | INFO     | zed_eval.docker:commit_image:282 - ✓ Committed: zed-eval-session:QuitmanWakeley_test__16f368e6_3_fix_order

✓ TURN 3 COMMITTED

Image: zed-eval-session:QuitmanWakeley_test__16f368e6_3_fix_order

TURN 4 IN PROGRESS
```